

# Backtesting Guide

AI Forex Trading System

*EURUSD M5 Scalper*

June 2026

## 1 Running a Backtest

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The quickest way to run a backtest:

```
python3 trade.py backtest
```

This runs a cumulative backtest from May 6 to Jun 5 with the current config settings (session hours, risk params, etc.).

### Custom Date Range

To backtest a specific period:

```
python3 trade.py backtest --from-date 2026-05-06
```

You can also use the raw runner for more control:

```
python3 run_backtest.py \  
  --csv data/raw/EURUSD_M5_60d.csv \  
  --symbol EURUSD \  
  --from-date 2026-05-06 \  
  --balance 100
```

### Single Day Backtest

```
python3 run_backtest.py \  
  --csv data/raw/EURUSD_M5_60d.csv \  
  --symbol EURUSD \  
  --date 2026-05-15 \  
  --balance 100
```

## 2 Reading Backtest Results

Example output:

```
Results: 83 trades (83 closed)
Performance Metrics
=====
Trade Period:      2026-05-06 -> 2026-06-05
Total Trades:      83
Wins / Losses:     23 / 60
Win Rate:          27.7%
Profit Factor:      1.53
Total P&L:          $+94.34
Final Balance:     $194.34
Avg Win / Loss:    $+11.83 / $-2.96
Max Drawdown:      $22.45 (10.6%)
Sharpe Ratio:      6.4
=====
```

### What Each Metric Means

- Win Rate: Percentage of trades that hit TP. Target 30-40% for 1:4 RR.
- Profit Factor: Gross profit / gross loss. Above 1.5 is good, above 2.0 is excellent.
- Final Balance: Starting \$100 + total P&L. Measures absolute return.
- Max Drawdown: Largest peak-to-trough decline. Keep under 15%.
- Sharpe Ratio: Risk-adjusted return. Above 3 is excellent for forex.
- Avg Win / Loss: Average winning and losing trade in dollars.

### The 1:4 RR Rule

With 4:1 reward-to-risk, the break-even win rate is 20%.

At 27.7% WR, expected value per trade =  $(0.277 \times 4) - (0.723 \times 1) = +0.39$

The system is profitable as long as WR stays above 20%.

### 3 Comparing Session Hours

The session hours config setting has a big impact on results.

| Session   | Trades | WR    | PF   | Final Bal | Max DD | Sharpe | Hours (Manila) |
|-----------|--------|-------|------|-----------|--------|--------|----------------|
| 03-10 UTC | 53     | 37.7% | 2.87 | \$281     | 5.9%   | 15.2   | 11AM-6PM       |
| 03-16 UTC | 83     | 27.7% | 1.53 | \$194     | 10.6%  | 6.4    | 11AM-12AM      |

The 03-10 UTC session (11AM-6PM Manila) captures London open and early NY overlap. Adding 10-16 UTC (6PM-12AM Manila) dilutes quality -- the system's PA patterns work best in the earlier session.

#### How to Change Session Hours

Edit config.py:

```
TRADE_SESSION_START = 3      # 03:00 UTC = 11:00 Manila
TRADE_SESSION_END   = 10     # 10:00 UTC = 18:00 Manila
```

Manila time = UTC + 8. To convert: add 8 to UTC hour.

```
Manila 11AM = 03 UTC   | Manila 6PM = 10 UTC
Manila 8PM  = 12 UTC   | Manila 12AM = 16 UTC
```

## 4 Changing Other Settings

All tunable parameters are in config.py:

```

BUY_THRESHOLD / SELL_THRESHOLD      # Signal strength required
ML_CONFIDENCE_GATE                 # ML prob gate (0.15 = need >0.65)
BASE_RISK_PCT                      # Risk per trade (0.015 = 1.5%)
ATR_MULT: sl: 1.0, tp: 4.0         # SL/TP as ATR multiples
MAX_DAILY_TRADES                   # Daily trade limit
MAX_DAILY_LOSS_PCT                 # Daily loss cap (0.03 = 3%)
W_PA, W_ML, W_SENT                 # Signal fusion weights
XGB_N_ESTIMATORS                   # ML model complexity

```

### Workflow: Change -> Test -> Compare

1. Edit config.py (change one setting at a time)
2. Run: `python3 trade.py backtest`
3. Note the metrics
4. Repeat with different values
5. Compare results side by side

### Common Tweaks to Try

- Increase TP multiplier: `ATR_MULT tp: 4.0 -> 5.0` (higher RR, lower WR)
- Reduce risk: `BASE_RISK_PCT 0.015 -> 0.01` (smaller positions, lower DD)
- Tighter gate: `ML_CONFIDENCE_GATE 0.15 -> 0.20` (fewer trades, higher WR)
- Session window: Adjust `TRADE_SESSION_START/END` (test different hours)

## 5 ML Model Testing

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### Retrain the Model

```
python3 trade.py train
```

Retrains XGBoost on the most recent EURUSD M5 CSV in data/raw/.

The model predicts if price will move 8+ pips in the next 10 min.

Current accuracy: 57.8%. Training takes ~30 seconds.

### Walk-Forward Validation

Most rigorous test -- no look-ahead bias. Trains on past 30 days,

tests on next 5 days, slides forward.

```
python3 walk_forward.py
```

This runs 7+ independent test windows. All windows profitable in

the last run confirms the model generalizes to unseen data.

### Grid Search (Finding Optimal Params)

To find the best model settings:

```
python3 grid_search.py
```

Tests 25 label combinations (lookahead x threshold) and 28

hyperparameter groups. Takes ~5 minutes. Results show best

accuracy, feature importance, and optimal config.

## 6 Quick Command Reference

```
# Daily use
python3 trade.py           Interactive menu
python3 trade.py live      Start live monitor
python3 trade.py status    System health check
python3 trade.py daily     Full daily routine

# Backtesting
python3 trade.py backtest  Quick backtest (May 6 - Jun 5)
python3 run_backtest.py    Full control (see --help)
    --csv DATA_FILE
    --symbol EURUSD
    --date YYYY-MM-DD      Single day
    --from-date YYYY-MM-DD Cumulative from date
    --balance 100          Starting balance

# Model & Data
python3 trade.py train     Retrain XGBoost model
python3 trade.py download  Refresh EURUSD M5 data
python3 walk_forward.py    Walk-forward validation
python3 grid_search.py     Hyperparameter tuning

# Config
config.py                  All settings in one file
logs/backtest_eurusd.csv   Last trade log
```

### Output Files

```
logs/system.log           Full event log
logs/backtest_eurusd.csv  Last backtest trades
models/eurusd_xgb.pkl     Trained ML model
data/raw/EURUSD_M5_*.csv  Market data
```